

LLM training and validation loss

The "Verdict" short story $\approx 20,000$ characters
 $\approx 50,000$ tokens

→ Tokens → Byte Pair Encoding
→ subword

Divide dataset into training and validation
train_ratio = 0.90

train_data = text_data[0:90] # 90% first
val_data = text_data[90:] # Last 10%

→ Use DataLoader to chunk training and validation data into input and output datasets

context_size = 4
X₁ ← "I had always thought"
X₁ ← "had always thought Jack"

if stride = 1, X₂ = "had always thought Jack"

X₁: [I had always thought
Jack Gilman rather or
Chop genius -- trash] → input

Usually, in GPT, context_length =
stride = 4

LLM model

Y₁: [had always thought Jack
Gilman rather or Chop
genius -- trash or]

Loss

LLM output

Let's look at first input: "I had always thought" → RNN model

Target!

"I had always thought Jack"

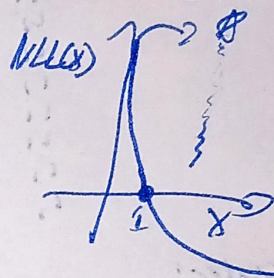
Labels

I	0.00	0.00	0.00	0.00
had	0.00	0.00	0.00	0.00
always	0.00	0.00	0.00	0.00
thought	0.00	0.00	0.00	0.00

Get output tokens
for target tokens

vocab-size = 50257

softmax
Probabilities



p_1
p_2
p_3
p_4

Cross-entropy loss

$$-\frac{1}{4} [\log p_1 + \log p_2 + \log p_3 + \log p_4]$$

Target!

had	23
always	3881
thought	11223
Jack	15

Token IDs

Now, look at how one batch is processed.

$x_1 =$ "I had always thought
Jack Gibson rather or"

$x_2 =$ "I had always thought Jack
Gibson rather or Cheap"

MBJH
PAGE

Labels

I	0.00	0.00	0.00	0.00
had	0.00	0.00	0.00	0.00
always	0.00	0.00	0.00	0.00
thought	0.00	0.00	0.00	0.00
Jack	0.00	0.00	0.00	0.00
Gibson	0.00	0.00	0.00	0.00
rather	0.00	0.00	0.00	0.00
or	0.00	0.00	0.00	0.00

losses.flatten(0,1)

I	0.00	0.00	0.00	0.00
had	0.00	0.00	0.00	0.00
always	0.00	0.00	0.00	0.00
thought	0.00	0.00	0.00	0.00
Jack	0.00	0.00	0.00	0.00
Gibson	0.00	0.00	0.00	0.00
rather	0.00	0.00	0.00	0.00
or	0.00	0.00	0.00	0.00

8x50257

vocab-size =

50257

284x50257

Target:

head
always
thought
Jaal
Gibson
weather
9
Lcheap

2

256

Play
page

123
3581
11223
15
8771
31412
148
9915

Get output tokens from target tokens

P₁₁
P₁₂
P₁₃
P₁₄
P₂₁
P₂₂
P₂₃
P₂₄

Cross entropy
Loss

$$= \frac{1}{4} \sum_{i=1}^8 \log(p_{i,j})$$

Train loader

torch.size(2, 256)

11
11
11
11
11
11
11

torch.size(2, 256)

11
11
11
11
11
11
11

9 training set batches
with 2 samples,
256 tokens each

Validation loader:

torch.size(2, 256) torch.size(2, 256)

→ one validation batch w/
2 samples, 256 tokens
each